

Physical activity and immune function in elderly women.

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The relationship between cardiorespiratory exercise, immune function, and upper respiratory tract infection (URTI) was studied in elderly women utilizing a randomized controlled experimental design with a follow-up of 12 wk. Thirty-two sedentary, elderly Caucasian women, 67-85 yr of age, who met specific selection criteria, were randomized to either a walking or calisthenic group; 30 completed the study. Twelve highly conditioned elderly women, 65-84 yr of age, who were active in endurance competitions, were recruited at baseline for cross-sectional comparisons. Intervention groups exercised 30-40 min, 5 d.wk⁻¹, for 12 wk, with the walking group training at 60% heart rate reserve and the calisthenic group engaging in mild range-of-motion and flexibility movements that kept their heart rates close to resting levels. At baseline, the highly conditioned subjects exhibited superior NK (119 +/- 13 vs 77 +/- 8 lytic units, $P < 0.01$) and T (33.3 +/- 4.9 vs 21.4 +/- 2.1 cpm $\times 10^{-3}$) using PHA, $P < 0.05$) cell function, despite no differences in circulating levels of lymphocyte subpopulations. Twelve weeks of moderate cardiorespiratory exercise improved the VO₂max of the sedentary subjects 12.6%, but did not result in any improvement in NK cell activity or T cell function. Incidence of URTI was lowest in the highly conditioned group and highest in the calisthenic control group during the 12-wk study, with the walkers in an intermediate position (chi-square = 6.36, $P = 0.042$). In conclusion, the highly conditioned elderly women in this study had superior NK and T cell function when compared with their sedentary counterparts.(ABSTRACT TRUNCATED AT 250 WORDS)